

Multi-step Estimators and Shrinkage Effect in Time Series Models

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IWSM June 2026

We all have our favourite time series models (do we?)

ETS, ARIMA, GARCH, STAR, ...

We know how to estimate them.

We have our favourite loss function.

How do you estimate your model?

Typically, it is a likelihood maximisation.

In case of Normal distribution, MLE corresponds to OLS.

OLS is equivalent to minimisation of Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T e_{t+1|t}^2$$

where $e_{t+1|t} = y_{t+1} - \hat{y}_{t+1|t}$ is in-sample forecast error.

When we are interested in h steps ahead forecast, we can use a different approach.

Produce h steps ahead in sample forecasts and minimise:

$$\text{MSE}_h = \frac{1}{T-h} \sum_{t=1}^T e_{t+h|t}^2$$

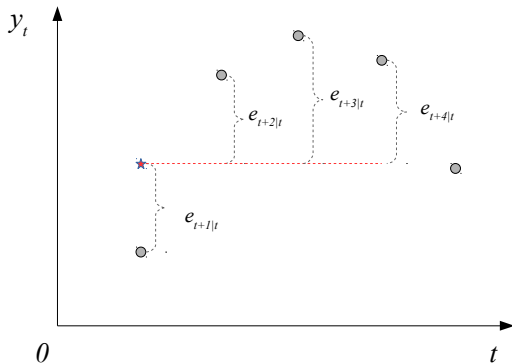
where $e_{t+h|t} = y_{t+h} - \hat{y}_{t+h|t}$.

MSE_h can be used for different h (Kang, 2003; Chevillon and Hendry, 2005; Pesaran et al., 2010)

This leads to h different values of parameters for each $j = 1, \dots, h$.

This is also called “direct multi-step estimator” (DMS, Chevillon, 2007).

Multistep losses imply producing h steps ahead forecasts from each t :



MSE_h produces estimates of parameters that are less efficient than with MSE.

(Tiao and Xu, 1993)

But it is more robust and increases forecast accuracy of models.

(McElroy and Wildi, 2013; Tiao and Xu, 1993; Clements and Hendry, 1996)

Although Marcellino et al. (2006) demonstrate that the forecast accuracy using MSE_h is lower than using MSE.

Estimation process of MSE_h is complex (h models to fit).

Weiss and Andersen (1984); Xia and Tong (2011) proposed using “Trace Mean Squared Error” (TMSE):

$$\text{TMSE} = \sum_{j=1}^h \frac{1}{T-h} \sum_{t=1}^T e_{t+j|t}^2.$$

We only fit model once, but estimates of parameters will have some benefits from MSE_h .

Kourentzes and Trapero (2018) show that the losses in accuracy in TMSE in comparison with MSE_h are marginal.

One more multistep estimator.

Kourentzes et al. (2019) proposed using “Mean Squared Cumulative Error” (MSCE):

$$\text{MSCE} = \frac{1}{T-h} \sum_{t=1}^T \left(\sum_{j=1}^h e_{t+j|t} \right)^2.$$

The motivation is to align this loss with typical inventory decisions.

We might be interested in cumulative demand over the lead time h .

Kourentzes et al. (2019) showed that it works well in context of inventory control.

So, what's the problem?

- The results are ambiguous;
- Estimates of parameters are inefficient;
- Estimates of parameters could be unstable;
- Nobody has ever explained why multistep estimators work / don't work;

There is no comprehensible explanation of what happens with models, when multistep estimators are used.

MSE is proportional to variance of one-step-ahead error.

If the forecasts are unbiased then:

$$\text{MSE}_h = \hat{\sigma}_h^2,$$

$$\text{TMSE} = \sum_{j=1}^h \sigma_j^2,$$

$$\text{MSCE} = \sum_{j=1}^h \sigma_j^2 + 2 \sum_{j=2}^h \sum_{i=1}^{j-1} \sigma_{i,j},$$

where $\sigma_{i,j}$ is the covariance between i^{th} and j^{th} steps ahead forecast errors.

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General form of pure additive ETS model (Snyder, 1985; Ord et al., 1997):

$$\begin{cases} y_t = \mathbf{w}'\mathbf{v}_{t-1} + \varepsilon_t \\ \mathbf{v}_t = \mathbf{F}\mathbf{v}_{t-1} + \mathbf{g}\varepsilon_t \end{cases},$$

see Svetunkov et al. (2024) for detail.

The h steps ahead variance of this model is:

$$\sigma_h^2 = \begin{cases} \sigma_1^2 \left(1 + \sum_{j=1}^{h-1} c_j^2 \right) & \text{when } h > 1, \\ \sigma_1^2 & \text{when } h = 1 \end{cases},$$

where

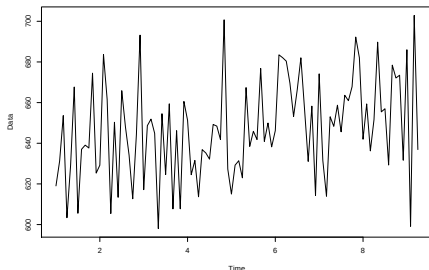
$$c_j = \mathbf{w}' \mathbf{F}^{j-1} \mathbf{g}.$$

When we use multistep estimators, we minimise $\hat{\sigma}_1^2$ together with $\hat{c}_j$.

An example with local level model, ETS(A,N,N):

$$\begin{aligned}y_t &= l_{t-1} + \varepsilon_t \\l_t &= l_{t-1} + \alpha \varepsilon_t\end{aligned}$$

so that $\mathbf{F} = 1$, $\mathbf{w} = 1$, $\mathbf{g} = \alpha$ and $\mathbf{v}_t = l_t$.





Values for ETS(A,N,N)

$$\hat{c}_j = \hat{\alpha}$$

$$\hat{\sigma}_h^2 = \hat{\sigma}_1^2 \left(1 + (h-1)\hat{\alpha}^2 \right)$$

So for this model:

$$\text{MSE}_h = \hat{\sigma}_1^2 \left(1 + (h-1)\hat{\alpha}^2 \right)$$

If we use multistep estimators, **we shrink smoothing parameter $\hat{\alpha}$** towards zero.

ETS(A,N,N) with $\hat{\alpha} = 0$ is a global level model (deterministic).



In case of TMSE, the shrinkage will be weaker for the same h :

$$\begin{aligned}\text{TMSE} &= \sum_{j=1}^h \hat{\sigma}_j^2 = \hat{\sigma}_1^2 \left(h + \sum_{j=2}^h \sum_{i=1}^{j-1} \hat{c}_{|j-i|}^2 \right) \\ &= \hat{\sigma}_1^2 h \left(1 + \frac{h-1}{2} \hat{\alpha}^2 \right)\end{aligned}$$

In MSCE, it will be potentially stronger than in TMSE.

Shrinkage is different for different models.

Model	$\hat{c}_j$	$\hat{\sigma}_h^2$
ETS(A,N,N)	$\hat{\alpha}$	$\hat{\sigma}_1^2 (1 + (h-1)\hat{\alpha}^2)$
ETS(A,A,N)	$\hat{\alpha} + \hat{\beta}j$	$\hat{\sigma}_1^2 \left(1 + \sum_{j=1}^{h-1} (\hat{\alpha} + \hat{\beta}j)^2\right)$
ETS(A,Ad,N)	$\hat{\alpha} + \hat{\beta} \sum_{i=1}^j \hat{\phi}^i$	$\hat{\sigma}_1^2 \left(1 + \sum_{j=1}^{h-1} \left(\hat{\alpha} + \hat{\beta} \sum_{i=1}^j \hat{\phi}^i\right)^2\right)$
ETS(A,N,A)	$\hat{\alpha} + \hat{\gamma}j_m$	$\hat{\sigma}_1^2 \left(1 + \sum_{j=1}^{h-1} (\hat{\alpha} + \hat{\gamma}j_m)^2\right)$

Table: Parameters of ETS models (Svetunkov et al., 2024). $j_m = \lfloor \frac{j-1}{m} \rfloor$, where m is seasonal periodicity.

Subconclusions:

1. Multistep estimators impose shrinkage on parameters;
2. This shrinkage makes models deterministic;
3. Strength of shrinkage is proportional to forecast horizon h ;
4. Shrinkage effect will differ from one model to another.

What about other models?

ARIMA can be represented in state space form (Snyder, 1985). First expand it:

$$y_t = \sum_{j=1}^K \varphi_j B^j y_{t-j} + \sum_{j=1}^K \eta_j B^j \varepsilon_{t-j} + \beta + \varepsilon_t.$$

Then:

$$\mathbf{w} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \mathbf{F} = \begin{pmatrix} \varphi_1 & 1 & 0 & \dots & 0 & 1 \\ \varphi_2 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \varphi_K & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{pmatrix}, \mathbf{g} = \begin{pmatrix} \varphi_1 + \eta_1 \\ \varphi_2 + \eta_2 \\ \vdots \\ \varphi_K + \eta_K \\ 0 \end{pmatrix}.$$

Using the same principles, we can have multiple steps ahead variance.

But it cannot be analysed in general... So, examples!

ARIMA(1,1,1):

$$\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{F} = \begin{pmatrix} 1 + \phi_1 & 1 \\ -\phi_1 & 0 \end{pmatrix}, \mathbf{g} = \begin{pmatrix} 1 + \phi_1 + \theta_1 \\ -\phi_1 \end{pmatrix}.$$

$$\hat{c}_j = 1 + \left(\hat{\phi}_1 + \hat{\theta}_1 \right) \sum_{i=1}^j \hat{\phi}_1^{i-1}.$$

Substituting into the formula of h steps ahead variance results in the following:

$$\hat{\sigma}_h^2 = \hat{\sigma}_1^2 \left(1 + \sum_{j=1}^{h-1} \left(1 + \left(\hat{\phi}_1 + \hat{\theta}_1 \right) \sum_{i=1}^j \hat{\phi}_1^{i-1} \right)^2 \right).$$

What does this mean?

We deal with geometric progression, thus:

$$\left(1 + (\hat{\phi}_1 + \hat{\theta}_1) \sum_{i=1}^j \hat{\phi}_1^{i-1}\right)^2 = \left(\frac{\hat{\theta}_1(1 - \hat{\phi}_1^j) + 1 - \hat{\phi}_1^{j+1}}{1 - \hat{\phi}_1}\right)^2,$$

This ratio will be minimised, when:

- $\hat{\phi}_1 \rightarrow -1$;
- $\hat{\theta}_1(1 - \hat{\phi}_1^j) + 1 - \hat{\phi}_1^{j+1} \rightarrow 0$;

MSE_h implies: $\hat{\theta}_1 \rightarrow -\frac{1 - \hat{\phi}_1^{j+1}}{1 - \hat{\phi}_1^j}$.

Together with MSE, this means: $\hat{\theta}_1 \rightarrow -1$.

In ARIMA(1,1,1) context it means that this model:

$$(1 - B)(1 - \hat{\phi}_1 B)y_t = (1 + \hat{\theta}_1 B)e_t$$

will become:

$$(1 - B)(1 + B)y_t = (1 - B)e_t,$$

or equivalently:

$$(1 + B)y_t = e_t,$$

which is a non-stationary ARIMA(1,0,0).

Other ARIMA models can be analysed in a similar way.

Subconclusions:

1. Shrinkage in ARIMA leads to degenerate models;
2. AR and MA parameters would shrink towards bounds;
3. These are typically "deterministic" models.

If you need to analyse parameters of ARIMA, be careful, when using multistep estimators.

Is it possible to weaken shrinkage, but still have benefits of multistep estimators?

The variance h steps ahead will be typically larger than 1 step ahead.

MSE_h will have strong shrinkage because of that.

TMSE will put emphasise on errors h steps ahead.

A possible solution – “Geometric TMSE” (GTMSE):

$$\text{GTMSE} = \sum_{j=1}^h \log \left(\frac{1}{T-h} \sum_{t=1}^{T-h} e_{t+j}^2 \right).$$

For the state space model we will have:

$$\text{GTMSE} = h \log \left(\hat{\sigma}_1^2 \right) + \sum_{j=2}^h \log \left(1 + \sum_{i=1}^{j-1} \hat{c}_i^2 \right).$$

Due to logarithms, the one-step-ahead variance is balanced out with the sum of $\hat{c}_j$ elements.

There is a shrinkage, but it is reduced in comparison with MSE_h , TMSE or MSCE .

Two DGPs:

1. ETS(A,N,N) with $\alpha = 0.2$;
2. ARIMA(0,1,1) with $\alpha = 0.6$;

500 time series for each

5000 observations

Get subsamples of 20, 50, 100, 200, 500, 1000 and 5000 observations.

Apply several models:

1. ETS(A,N,N);
2. ETS(A,A,N);
3. ARIMA(0,1,1);
4. ARIMA(1,1,1);

This covers:

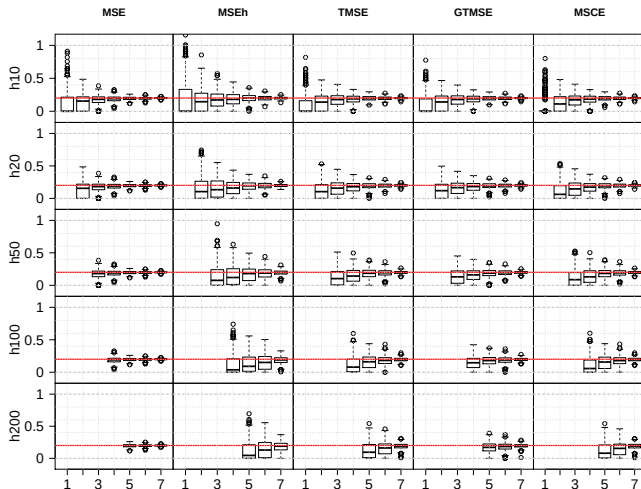
- The model is correctly specified;
- The model is analogous to the true one (e.g., ETS(A,N,N) and ARIMA(0,1,1));
- The model is misspecified (e.g., ARIMA(1,1,1) and ARIMA(0,1,1));
- The model is wrong (e.g., ETS(A,A,N) and ARIMA(0,1,1)).

Use all the discussed estimators:

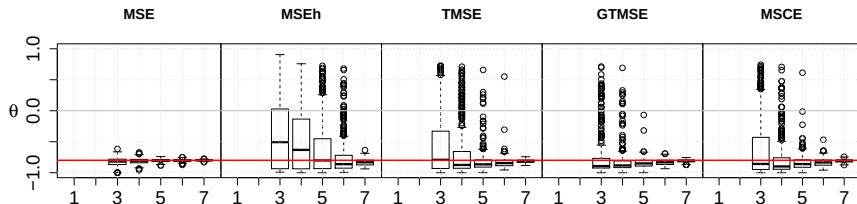
1. MSE;
2. MSE_h ;
3. TMSE;
4. GTMSE;
5. MSCE.

Record estimated parameters.

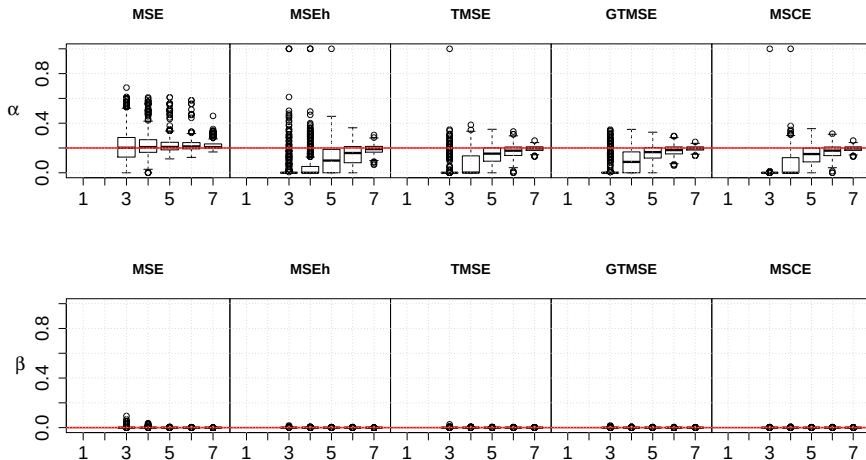
Use `es()`, `ssarima()`, `sim.es()`, `sim.ssarima()` functions from `smooth v2.5.3` package (Svetunkov, 2019) for R.



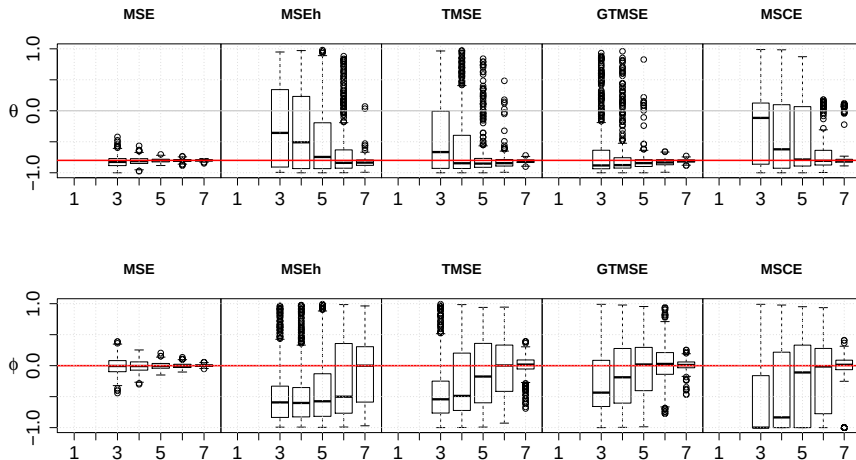
ARIMA(0,1,1) $h=50$



ETS(A,N,N) $h=50$



ARIMA(1,1,1) $h=50$



Summarising the plots above (and those that are not shown here):

- Shrinkage is the strongest in MSCE, followed by MSE_h ;
- GTMSE has the mildest shrinkage;
- It helps in case of misspecified model;
- Estimates of parameters are biased but can be more efficient than in case of MSE;
- Shrinkage is neutralised, when sample size increases.

Why bother with multistep estimators then?

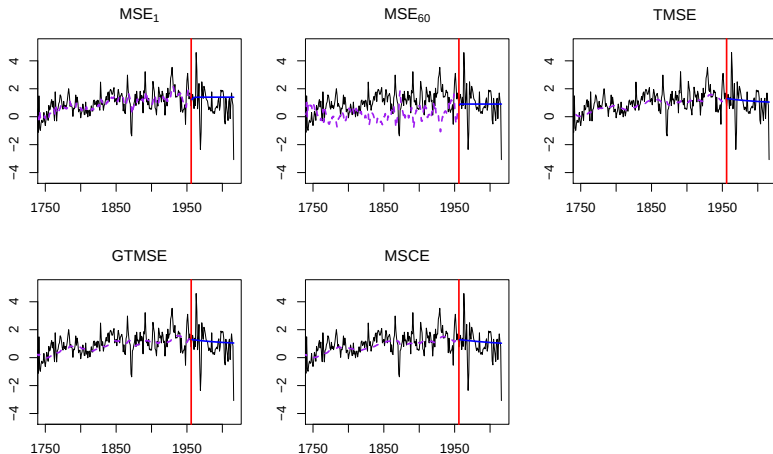
Hadley Centre / Climatic Research Unit data of Mean Surface Temperatures.

2016 observations, withholding 60 observations.

Apply ARIMA(1,1,2) (see Beaulieu and Killick, 2018, and discussion within) with:

1. MSE,
2. MSE_h ,
3. TMSE,
4. GTMSE,
5. and MSCE.

Example application



Example application

Estimators	RMSE	MAE	ME	AR(1)	MA(1)	MA(2)
MSE_1	1.216	0.848	-26.919	0.569	-1.248	0.277
MSE_h	1.129	0.785	2.342	-0.200	-1.311	0.315
TMSE	1.131	0.768	-12.470	0.983	-1.924	0.924
GTMSE	1.130	0.768	-12.441	0.982	-1.921	0.921
MSCE	1.129	0.767	-12.277	0.989	-1.941	0.941

Table: Performance of estimators on an example of a time series.

TMSE, GTMSE and MSCE estimates of the AR(1) parameter are consistent with Hartmann et al. (2013); Cahill et al. (2015); Rahmstorf et al. (2017)

- Multistep estimators imply shrinkage of parameters;
- Parameters of ETS and ARIMA shrink differently but towards deterministic model;
- This gives robustness to models and help in long-term forecasting;
- If you want accuracy for a specific horizon, use MSE_h ;
- But! Parameters may overshrink when estimated using MSE_h and MSTFE;
- If you care for parameters, MSE is a safer choice;
- If you are unsure, GTMSE is a balanced estimator;

- Beaulieu, C., Killick, R., 2018. Distinguishing trends and shifts from memory in climate data. *Journal of Climate* 31 (23), 9519–9543.
- Cahill, N., Rahmstorf, S., Parnell, A. C., 2015. Change points of global temperature. *Environmental Research Letters* 10 (8), 084002.
- Chevillon, G., sep 2007. Direct Multi-Step Estimation and Forecasting. *Journal of Economic Surveys* 21 (4), 746–785.
URL <http://doi.wiley.com/10.1111/j.1467-6419.2007.00518.x>
- Chevillon, G., Hendry, D. F., apr 2005. Non-parametric direct multi-step estimation for forecasting economic processes. *International Journal of Forecasting* 21 (2), 201–218.
URL <http://linkinghub.elsevier.com/retrieve/pii/S016920700400069X>

Clements, M. P., Hendry, D. F., may 1996. Multi-Step Estimation for Forecasting. Oxford Bulletin of Economics and Statistics 58 (4), 657–684.

URL [http:](http://doi.wiley.com/10.1111/j.1468-0084.1996.mp58004005.x)

[//doi.wiley.com/10.1111/j.1468-0084.1996.mp58004005.x](http://doi.wiley.com/10.1111/j.1468-0084.1996.mp58004005.x)

Hartmann, D., Klein Tank, A., Rusticucci, M., Alexander, L., Brönnimann, S., Charabi, Y., Dentener, F., Dlugokencky, E., Easterling, D., Kaplan, A., Soden, B., Thorne, P., Wild, M., Zhai, P., 2013. Observations: Atmosphere and Surface. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, book section 2, pp. 159–254.

Kang, I.-B., 2003. Multi-period forecasting using different models for different horizons: an application to U.S. economic time series data. *International Journal of Forecasting* 19 (3), 387–400.

URL [http:](http://linkinghub.elsevier.com/retrieve/pii/S0169207002000109)

[//linkinghub.elsevier.com/retrieve/pii/S0169207002000109](http://linkinghub.elsevier.com/retrieve/pii/S0169207002000109)

Kourentzes, N., Li, D., Strauss, A. K., 2019. Unconstraining methods for revenue management systems under small demand. *Journal of Revenue and Pricing Management* 18 (1), 27–41.

Kourentzes, N., Trapero, J. R., 2018. On the use of multi-step cost functions for generating forecasts.

Marcellino, M., Stock, J. H., Watson, M. W., nov 2006. A comparison of direct and iterated multistep AR methods for forecasting macroeconomic time series. *Journal of Econometrics* 135 (1-2), 499–526.

URL [http:](http://linkinghub.elsevier.com/retrieve/pii/S030440760500165X)

[//linkinghub.elsevier.com/retrieve/pii/S030440760500165X](http://linkinghub.elsevier.com/retrieve/pii/S030440760500165X)

McElroy, T., Wildi, M., jul 2013. Multi-step-ahead estimation of time series models. *International Journal of Forecasting* 29 (3), 378–394.

URL <http://www.sciencedirect.com/science/article/pii/S0169207012001148>

- Ord, J. K., Koehler, A. B., Snyder, R. D., dec 1997. Estimation and Prediction for a Class of Dynamic Nonlinear Statistical Models. Journal of the American Statistical Association 92 (440), 1621–1629. URL <http://www.tandfonline.com/doi/abs/10.1080/01621459.1997.10473684>
- Pesaran, M. H., Pick, A., Timmermann, A., 2010. Variable Selection, Estimation and Inference for Multi-period Forecasting Problems.
- R Core Team, 2018. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org/>
- Rahmstorf, S., Foster, G., Cahill, N., apr 2017. Global temperature evolution: recent trends and some pitfalls. Environmental Research Letters 12 (5), 054001.

- Snyder, R. D., 1985. Recursive Estimation of Dynamic Linear Models. *Journal of the Royal Statistical Society, Series B (Methodological)* 47 (2), 272–276.
- Svetunkov, I., 2019. smooth: Forecasting Using State Space Models. R package version 2.5.3.
URL <https://github.com/config-i1/smooth>
- Svetunkov, I., Kourentzes, N., Killick, R., 2024. Multi-step estimators and shrinkage effect in time series models. *Computational Statistics* 39, 1203–1239.
- Tiao, G. C., Xu, D., 1993. Robustness of maximum likelihood estimates for multi-step predictions: the exponential smoothing case. *Biometrika* 80 (3), 623–641.
URL <http://biomet.oxfordjournals.org/content/80/3/623.short>

- Weiss, A. A., Andersen, A. P., sep 1984. Estimating Time Series Models Using the Relevant Forecast Evaluation Criterion. Journal of the Royal Statistical Society. Series A 147 (3), 484.
- Xia, Y., Tong, H., 2011. Feature matching in time series modeling. Statistical Science 26 (1), 21–46.
URL <http://www.imstat.org/sts/>